

Chapter 15 Homework - Eigenvalues, Eigenvectors, and Spectral Theory

Part B - Hand Calculation

Derive $|A - \lambda I| = 0$:

$$Av = \lambda v$$

$$Av - \lambda v = 0$$

$$v(A - \lambda I) = 0$$

For non-zero v ,

$$|A - \lambda I| = 0.$$

Plug in $A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$ to solve for eigenvalues and eigenvectors

$$\begin{vmatrix} 4-\lambda & 2 \\ 1 & 3-\lambda \end{vmatrix} = (4-\lambda)(3-\lambda) - 2 = \lambda^2 - 7\lambda + 10 = 0$$

We can factor the quadratic to get

$$(\lambda - 5)(\lambda - 2) = 0, \text{ so}$$

$$\boxed{\lambda_1 = 5, \lambda_2 = 2.}$$

For $\lambda_1 = 5$,

$$v(A - 5I) = 0$$

$$\begin{bmatrix} 4-5 & 2 \\ 1 & 3-5 \end{bmatrix} \cdot v = \begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix} \cdot v = 0$$

$$\boxed{v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.}$$

For $\lambda_2 = 2$,

$$v(A - 2I) = 0$$

$$\begin{bmatrix} 4-2 & 2 \\ 1 & 3-2 \end{bmatrix} \cdot v = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} \cdot v = 0$$

$$\boxed{v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.}$$

Q is just v_1 and v_2 stacked horizontally, so

$$\boxed{Q = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}.}$$

Λ is a diagonal matrix of the eigenvalues, so

$$\boxed{\Lambda = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}.}$$

We can solve for $Q^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ using two linear systems:

$$Q \cdot \begin{bmatrix} a \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } Q \cdot \begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

$$\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} a \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow 2a + c = 1, a - c = 0; \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow 2b + d = 0, b - d = 1$$

$$a = \frac{1}{3}, b = \frac{1}{3}, c = \frac{1}{3}, d = -\frac{2}{3}$$

$$\boxed{Q^{-1} = \frac{1}{3} \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}.}$$

Verify $A = Q\Lambda Q^{-1}$:

$$\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix} \cdot \frac{1}{3} = \begin{bmatrix} 10 & 2 \\ 5 & -2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix} \cdot \frac{1}{3} = \begin{bmatrix} 12 & 6 \\ 3 & 9 \end{bmatrix} \cdot \frac{1}{3} = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix} = A.$$